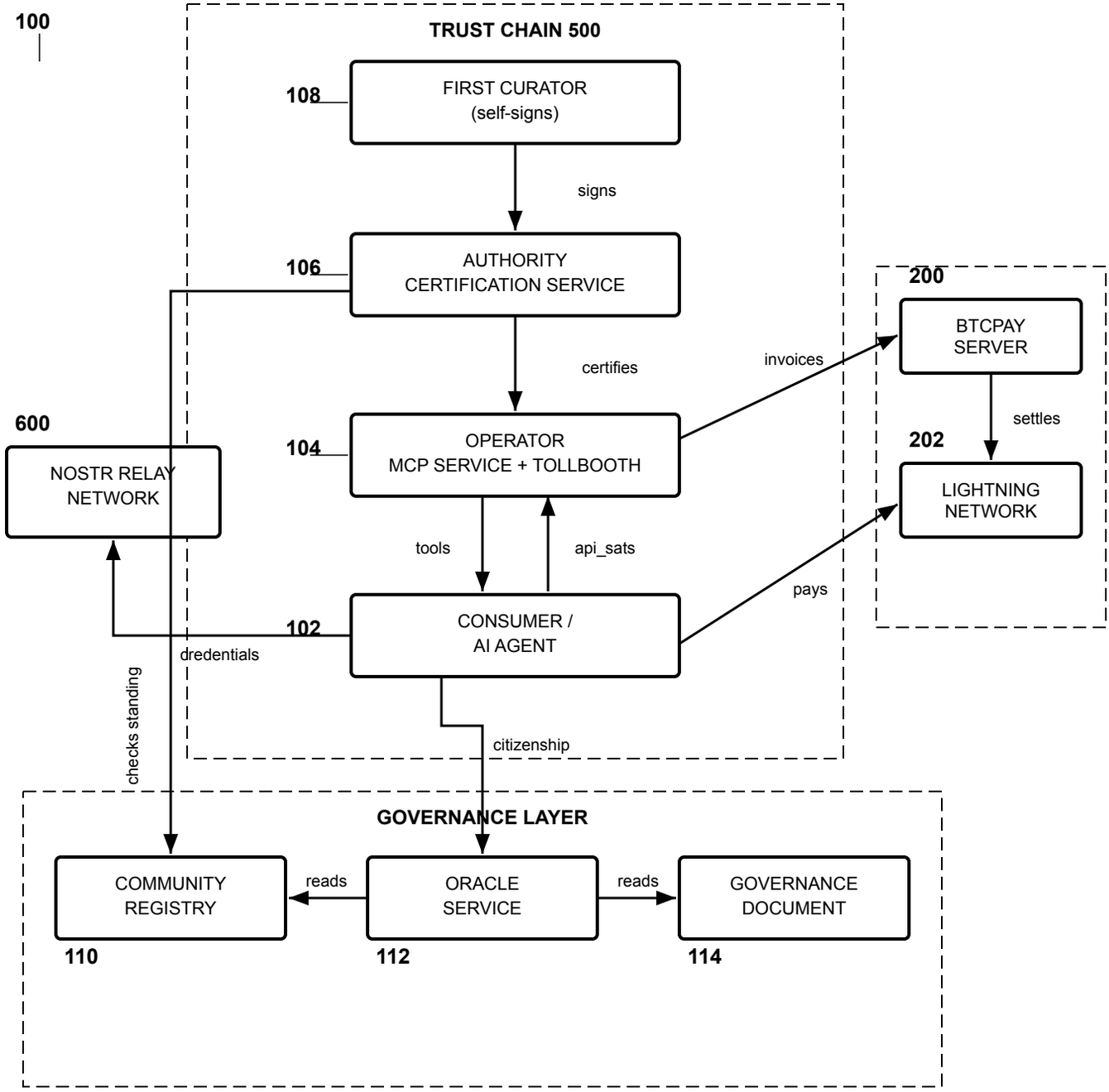

100
TRUST CHAIN 500
108
FIRST CURATOR
(self-signs)
signs
106
AUTHORITY
CERTIFICATION SERVICE
certifies
104
OPERATOR
MCP SERVICE + TOLLBOOTH
tools
api_sats
102
CONSUMER /
AI AGENT
dredentials
citizenship
600
NOSTR RELAY
NETWORK
checks standing
invoices
200
BTCPAY
SERVER
settles
202
LIGHTNING
NETWORK
pays
GOVERNANCE LAYER
COMMUNITY
REGISTRY
110
reads
ORACLE
SERVICE
112
reads
GOVERNANCE
DOCUMENT
114

FIG. 1

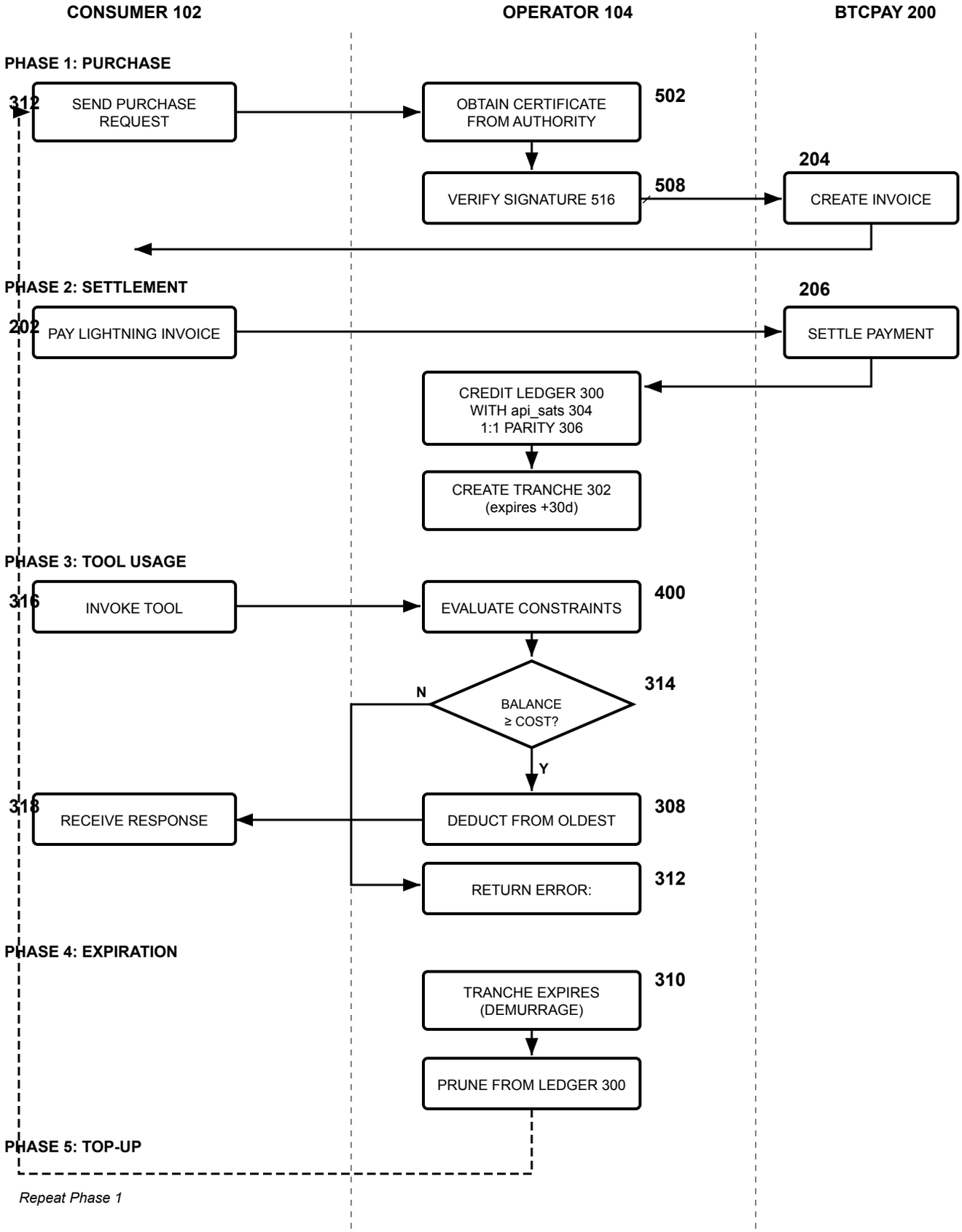

CONSUMER 102
OPERATOR 104
BTCPAY 200
PHASE 1: PURCHASE
312
SEND PURCHASE REQUEST
OBTAIN CERTIFICATE FROM AUTHORITY
502
VERIFY SIGNATURE 516
508
204
CREATE INVOICE
PHASE 2: SETTLEMENT
206
202
PAY LIGHTNING INVOICE
SETTLE PAYMENT
CREDIT LEDGER 300 WITH api_sats 304 1:1 PARITY 306
CREATE TRANCHE 302 (expires +30d)
PHASE 3: TOOL USAGE
316
INVOKE TOOL
EVALUATE CONSTRAINTS
400
N
BALANCE ≥ COST?
314
Y
318
RECEIVE RESPONSE
DEDUCT FROM OLDEST
308
RETURN ERROR:
312
PHASE 4: EXPIRATION
TRANCHE EXPIRES (DEMURRAGE)
310
PRUNE FROM LEDGER 300
PHASE 5: TOP-UP
Repeat Phase 1

FIG. 2

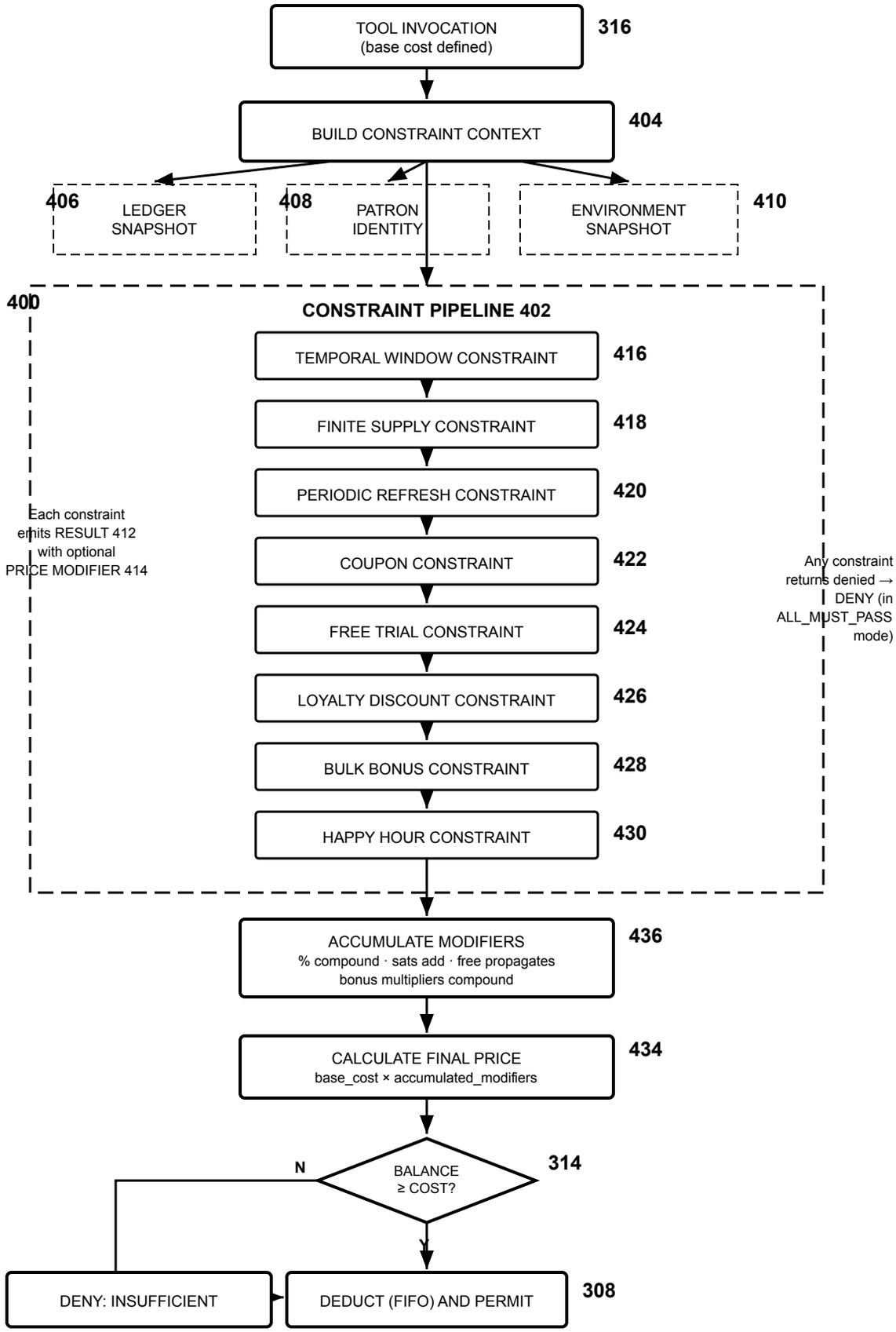

TOOL INVOCATION
(base cost defined)
316
BUILD CONSTRAINT CONTEXT
404
406
LEDGER SNAPSHOT
408
PATRON IDENTITY
ENVIRONMENT SNAPSHOT
410
400
CONSTRAINT PIPELINE 402
TEMPORAL WINDOW CONSTRAINT
416
FINITE SUPPLY CONSTRAINT
418
PERIODIC REFRESH CONSTRAINT
420
COUPON CONSTRAINT
422
FREE TRIAL CONSTRAINT
424
LOYALTY DISCOUNT CONSTRAINT
426
BULK BONUS CONSTRAINT
428
HAPPY HOUR CONSTRAINT
430
Each constraint emits RESULT 412 with optional PRICE MODIFIER 414
Any constraint returns denied → DENY (in ALL_MUST_PASS mode)
ACCUMULATE MODIFIERS
% compound · sats add · free propagates
bonus multipliers compound
436
CALCULATE FINAL PRICE
base_cost × accumulated_modifiers
434
BALANCE ≥ COST?
314
N
Y
DENY: INSUFFICIENT
DEDUCT (FIFO) AND PERMIT
308

FIG. 3

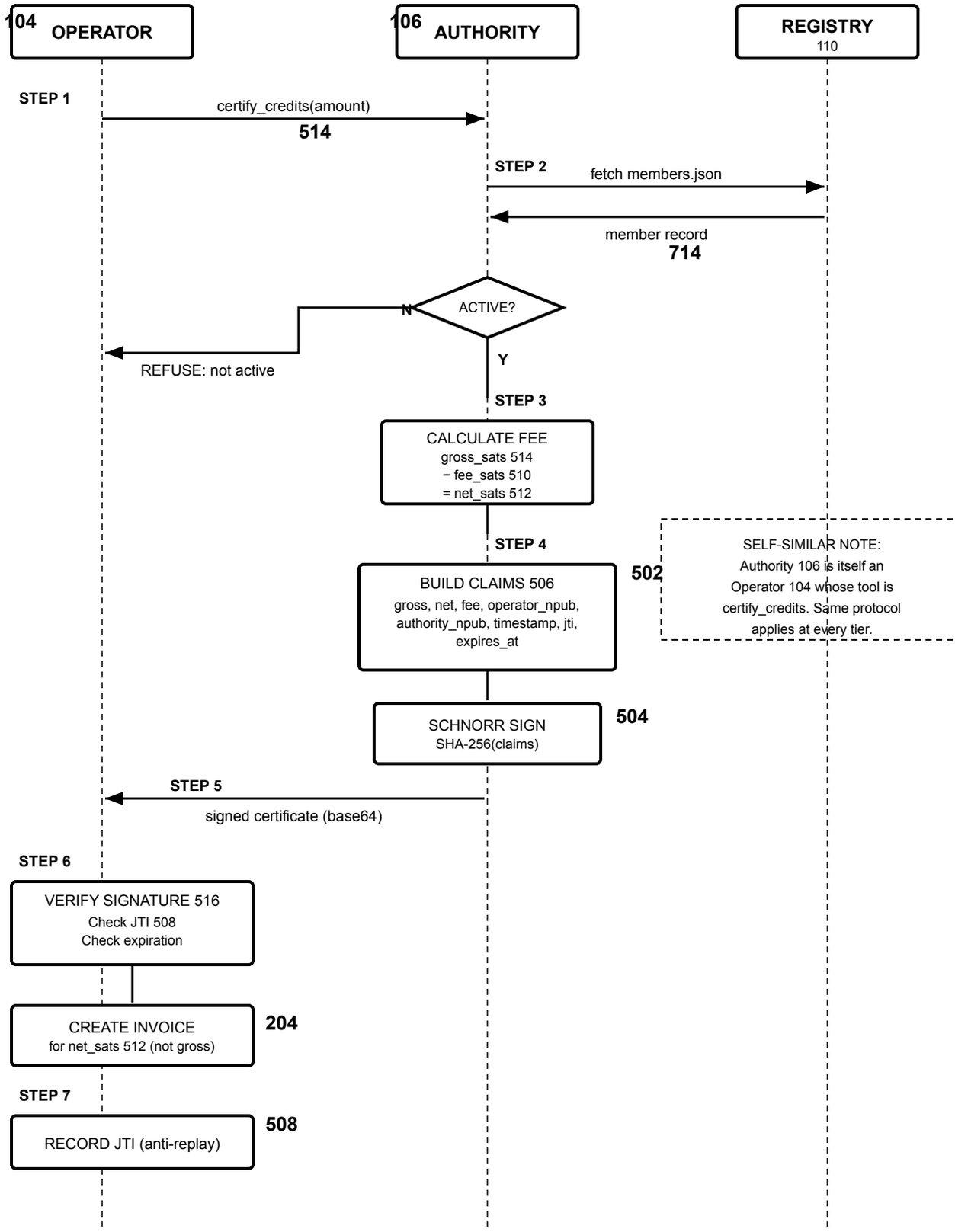

104
OPERATOR
106
AUTHORITY
REGISTRY
110
STEP 1
certify_credits(amount)
514
STEP 2
fetch members.json
member record
714
ACTIVE?
N
REFUSE: not active
Y
STEP 3
CALCULATE FEE
gross_sats 514
− fee_sats 510
= net_sats 512
STEP 4
BUILD CLAIMS 506
gross, net, fee, operator_npub, authority_npub, timestamp, jti, expires_at
502
SELF-SIMILAR NOTE:
Authority 106 is itself an Operator 104 whose tool is certify_credits. Same protocol applies at every tier.
SCHNORR SIGN
SHA-256(claims)
504
STEP 5
signed certificate (base64)
STEP 6
VERIFY SIGNATURE 516
Check JTI 508
Check expiration
CREATE INVOICE
for net_sats 512 (not gross)
204
STEP 7
RECORD JTI (anti-replay)
508

FIG. 4

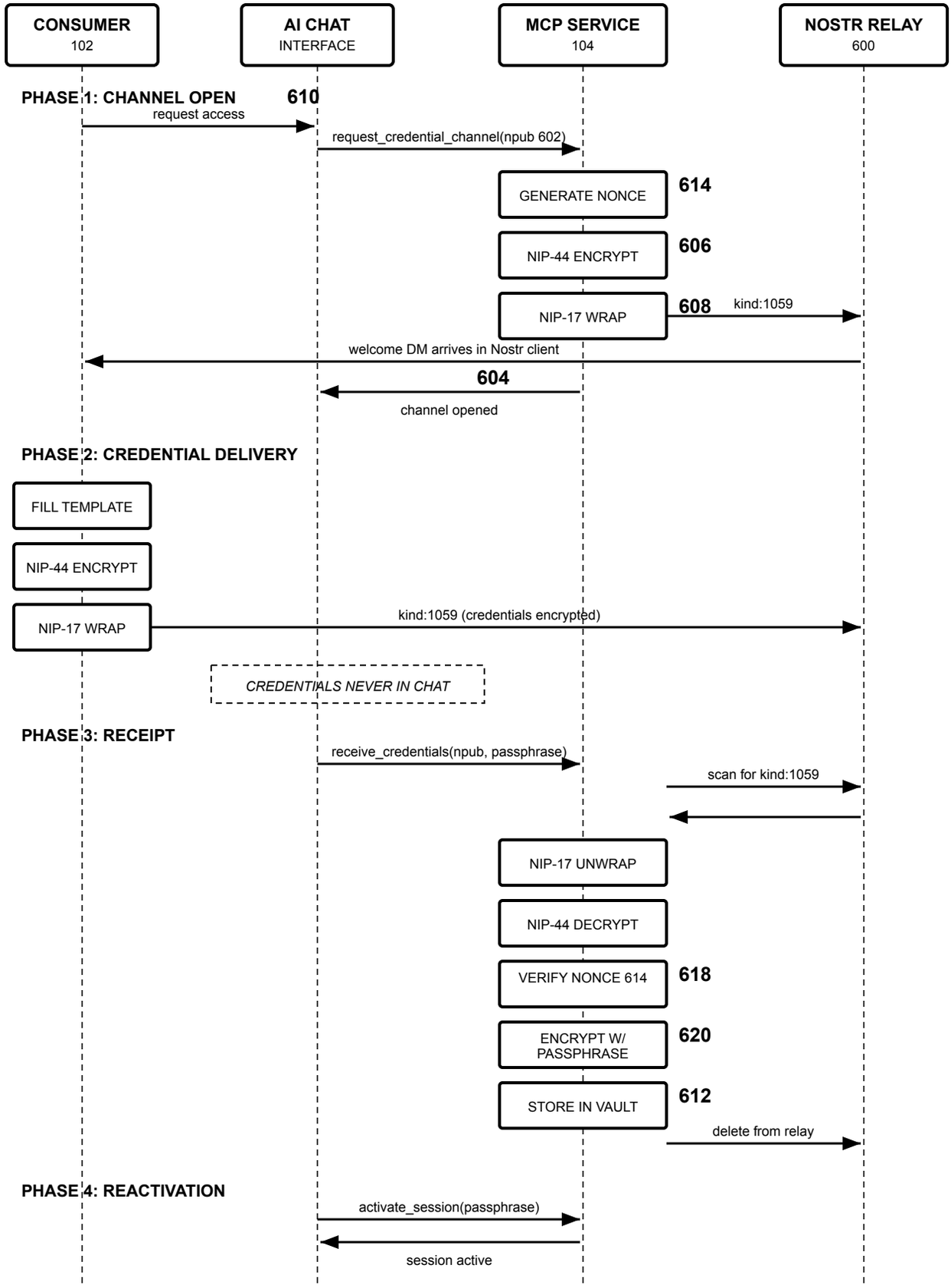
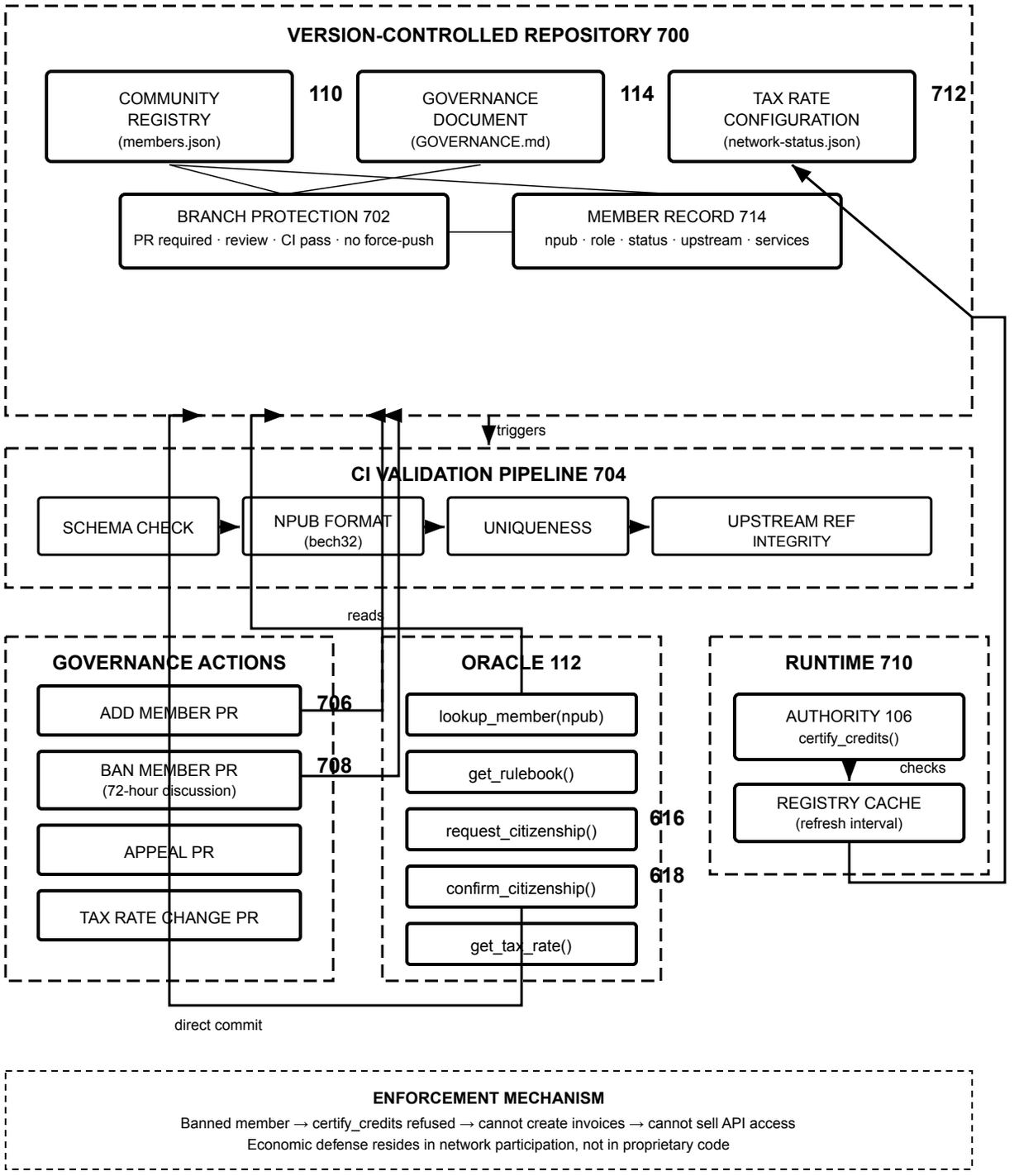

CONSUMER
102
AI CHAT
INTERFACE
MCP SERVICE
104
NOSTR RELAY
600
PHASE 1: CHANNEL OPEN
610
request access
request_credential_channel(npub 602)
GENERATE NONCE
614
NIP-44 ENCRYPT
606
NIP-17 WRAP
608
kind:1059
welcome DM arrives in Nostr client
604
channel opened
PHASE 2: CREDENTIAL DELIVERY
FILL TEMPLATE
NIP-44 ENCRYPT
NIP-17 WRAP
kind:1059 (credentials encrypted)
CREDENTIALS NEVER IN CHAT
PHASE 3: RECEIPT
receive_credentials(npub, passphrase)
scan for kind:1059
NIP-17 UNWRAP
NIP-44 DECRYPT
VERIFY NONCE 614
618
ENCRYPT W/ PASSPHRASE
620
STORE IN VAULT
612
delete from relay
PHASE 4: REACTIVATION
activate_session(passphrase)
session active

FIG. 5



ENFORCEMENT MECHANISM

Banned member → certify_credits refused → cannot create invoices → cannot sell API access

Economic defense resides in network participation, not in proprietary code

FIG. 6